Kuan-Hung Chen

+886 970 492 633 • kc.khchen@gmail.com • kckhchen • kckhchen

Education

University of Warwick — **Coventry, UK**
MSc in Statistics (*with Distinction*) — *Sep 2024 — Sep 2025*
Focus: *Machine Learning, Neural Networks, Statistical Inference, Model Evaluation*

National Taiwan University — **Taipei, Taiwan**
B.A. in Economics & B.A. in Foreign Languages and Literatures — *Sep 2019 — Jun 2024*
Double Major, Overall GPA: 3.9 / 4.3. Focus: *Econometrics, Game Theory*

Experience

Software System Designer — **Taipei, Taiwan**
Areté Inc. (*via Abacus Supply Chain Solution*) — *Oct 2026 — Present*
- Write technical specs for a supply chain forecasting and inventory planning SaaS, covering data model changes, calculation logic, and UI; review the implementation with engineers on GitLab and acceptance-test features against the spec before release.
- Specify the forecasting chatbot's anomaly-detection and remediation flow, and the API between its Python/FastAPI LLM service and the core Java platform.

Market Research Intern — **Taipei, Taiwan**
KANTAR Worldpanel — *Jun 2023 — Apr 2024*
- Engineered 10+ Python and VBA pipelines automating inspection, transformation, and reporting over consumer panel data, cutting report production time by over 20x; adopted across the analyst team.

Research and Projects

MSc Dissertation, University of Warwick
Probabilistic Weather Forecasting with Recursive Algorithms and Vine Copulas — *Sep 2025*
- Built a vine copula-based probabilistic forecasting model in PyTorch that absorbs new data through quasi-Bayesian recursive updates, without retraining.
- On real-world weather data, cut forecast error (CRPS) by 24% vs the strongest baseline, training in under 3 minutes on a laptop CPU.
- Co-authored a contributed talk extending this work to spatio-temporal forecasting, presented at the *Environmental Modelling meets Infectious Diseases* workshop, University of Warwick, Apr 2026.

Pollparse
ALBERT-based zh-TW poll parser powering Sootow (*open source*) — *Sep 2026*
- Fine-tuned an ALBERT encoder with PyTorch and Transformers to parse free-form zh-TW poll text into titles, options, settings, and time; built a held-out eval set of real user inputs, run as a regression gate in CI, reaching 87% exact match.
- Exported to ONNX and deployed as the backend API of Sootow, a live polling app, running a 16 MB model at about 1 ms per request on a single CPU core.
- Archived the app's traffic and usage metrics in PostgreSQL, with SQL views to tune review thresholds from real data.

zh-TW NLP Product Categorizer
TF-IDF + SVC/MNB classifier served with FastAPI and Docker (*open source*) — *Aug 2026*
- Combined character- and word-level TF-IDF features for mixed Chinese/English product names, classifying noisy real-world listings into 84 categories at 70% top-1 and 85% top-3 accuracy.

Open Source

Timeful — 1.7k-star Go/Vue scheduling platform
Security and reliability hardening (*event ID entropy, rate limiting*), *merged upstream* — *Sep 2026*
- Replaced predictable, time-seeded `math/rand` event IDs with `crypto/rand` and a wider alphabet, growing the keyspace from 3.2M to 24.8B so private event links can no longer be guessed (#274).
- Added per-IP rate limiting to unauthenticated lookup endpoints, charging only failed lookups (#275).

Skills

Programming: Python, Go, C++, SQL, JavaScript, Ruby, R

Machine Learning: PyTorch, Hugging Face Transformers, ONNX, scikit-learn, NLP, Model Evaluation, Time Series

Engineering: REST APIs, FastAPI, PostgreSQL, Docker, Linux, Git, CI/CD, Bash, Unit & Integration Testing